SESSION 2 — AFTERNOON

Macroeconomics

Business cycles, fiscal & monetary policy, trade and exchange rates

Session Roadmap — 2 Hours

Macroeconomics determines the environment in which all companies operate. These forces drive earnings, interest rates, exchange rates and policy risk.

01 Business Cycles

4 phases (CFA LM2) • Exhibit 5 characteristics table • credit cycles • asset class performance

02 Economic Indicators

Leading, coincident, lagging • US Leading Indicators table (CFA LM2) • PMI • yield curve

03 Fiscal Policy

Roles • Keynesian vs Monetarist • multiplier • government revenue/expenditure % GDP (CFA LM3)

04 Monetary Policy

Money supply • neutral rate • QE • monetary-fiscal combinations (CFA LM4)

05 International Trade

Comparative advantage • tariff analysis: South Africa paper (CFA LM6) • trade barriers

06 Exchange Rates

Spot rates • direct/indirect • USD/EUR +4.35% (CFA LM7) • MXN/USD=18.8590 • PPP

Learning Objectives

Business Cycle Analysis

Name and describe all 4 CFA business cycle phases. Read the CFA Exhibit 5 characteristics table for each phase.

Economic Indicators

Distinguish leading, coincident and lagging indicators. Identify the 10+ US leading indicators from CFA LM2 and their rationale.

Fiscal Policy

Describe fiscal policy tools. Distinguish Keynesian and Monetarist views. Interpret government revenue/expenditure data (CFA LM3).

Monetary Policy

Explain money creation, central bank tools, neutral rate and QE. Analyse fiscal-monetary policy combinations (CFA LM4).

International Trade

Apply comparative advantage. Calculate tariff effects: consumer surplus, producer surplus, government revenue and DWL using CFA South Africa example.

Exchange Rates

Compute currency appreciation/depreciation correctly (USD/EUR +4.35% from CFA LM7). Apply PPP and IRP frameworks.

SECTION 01

Business Cycles (CFA LM2)

GDP, Output Gaps & the Business Cycle

Output Gap = Actual GDP – Potential GDP **Positive gap = overheating** **Negative gap = recession**

The business cycle reflects fluctuations of actual GDP around its TREND (potential output). CFA uses the GROWTH CYCLE concept.

GDP Component	Formula (Expenditure Approach)	Typical % of GDP (Advanced Economy)
Consumer spending (C)	Household purchases of goods and services	55–60%
Investment (I)	Business capex + residential construction + inventory	15–20%
Government spending (G)	Public consumption and investment (not transfers)	15–20%
Net Exports (NX)	Exports – Imports	Varies (−5% to +5%)
GDP	C + I + G + NX	100%

Real vs Nominal GDP

Nominal GDP: output at CURRENT prices. Rises with inflation even if real output is unchanged.
Real GDP: output at CONSTANT base year prices. Removes inflation. Used for business cycle analysis.

GDP Deflator vs CPI

GDP Deflator = Nominal GDP / Real GDP × 100. Covers ALL domestically produced goods.
CPI covers a fixed basket of consumer goods. CPI more timely, used for monetary policy.

Four Phases of the Business Cycle — CFA LM2, Exhibit 5

CFA Vol 2 LM2 uses four named phases: Recovery, Expansion, Slowdown, Contraction. The cycle is measured relative to POTENTIAL output (growth cycle concept).

Phase	GDP Gap	Activity Levels	Inflation	Interest Rates	Asset Prices
Recovery	Large neg. gap, narrowing	Below potential but rising. Unemployment high. Consumer/business spending begins.	Low/falling	Low/stable. Central banks accommodative.	Equities begin to rise. B peak.
Expansion	Positive gap, widening	Above potential. Firms hire, invest, expand. Confidence rising.	Rising gradually	Rising. Central banks tighten.	Equities strong. Commo rise.
Slowdown	Positive gap, narrowing	Still above potential but decelerating. Firms use overtime. Confidence plateaus.	High/peaks	High, may plateau	Equities volatile. Bonds
Contraction	Negative gap, widening	Below potential. Unemployment rises. Firms cut costs, reduce headcount.	Falling	Falling. Central banks ease.	Equities weak. Bonds, g outperform.

Cycle Investing — Asset Class Rotation by Phase

CFA LM2: different sectors and asset classes outperform at different points in the cycle. Understanding the phase helps position portfolios.

Recovery	Equities (early cyclicals: consumer discretionary, financials, industrials). High yield credit spreads tighten.
Expansion	Equities (tech, materials, energy). Commodities. Inflation-linked bonds. Reduce duration on fixed income.
Slowdown	Quality equities (defensive sectors). Investment grade credit. Start increasing fixed income duration.
Contraction	Long-duration government bonds, gold, defensive equities (utilities, consumer staples, healthcare). Cash.

Caution: cycle rotation is a guide, not a formula. Turning points are only clear IN RETROSPECT. Markets often move 6–12 months ahead of the real economy (forward-looking).

Credit Cycles — CFA LM2: Amplifying the Business Cycle

CFA LM2 emphasises that credit cycles AMPLIFY business cycles. Credit expansion and contraction reinforce the economy's ups and downs.

Credit Expansion (Boom)

Banks compete to lend. Credit standards loosen. Asset prices rise. Collateral values increase, enabling more borrowing. Leverage builds.

Problems emerge: mispriced risk, overinvestment in unproductive assets, bubbles form in real estate/equities.

Fed/BIS research: credit-to-GDP gap is a reliable early warning of financial crises.

Credit Contraction (Bust)

Defaults rise. Banks tighten lending standards. Collateral values fall. Forced selling amplifies price declines. Credit crunch cuts off even creditworthy borrowers.

Deleveraging: simultaneous attempts to reduce debt. Deflationary if widespread. 2008-09 financial crisis is the textbook example.

Central banks must act as lender of last resort.

Credit Cycle vs Business Cycle

Credit cycles tend to be longer (8–20 years) and more extreme than business cycles (4–8 years). Recoveries after debt-driven contractions are typically slower (balance sheet recessions).
CFA states: 'A vulnerable financial system can amplify the effects of negative shocks, causing widespread recessions.'
The Leading Credit Index (part of the US Leading Indicators) captures financial system fragility.

SECTION 02

Economic Indicators (CFA LM2)

Leading, Coincident & Lagging Indicators

Economic indicators are classified by their TIMING relative to the business cycle. Analysts use them to identify the current phase and forecast turning points.

Type	Definition	Examples	Investment Use
Leading	Change BEFORE the economy turns	S&P 500, yield curve spread, building permits, ISM orders	Anticipate cycle turning points. Position 3–12 months ahead.
Coincident	Change AT THE SAME TIME as the economy	GDP, industrial production, non-farm payrolls, personal income	Confirm current phase. Useful for present-state assessment.
Lagging	Change AFTER the economy has turned	Unemployment rate, CPI, bank loan rates, inventory-to-sales ratio	Confirm the cycle has turned. Reduce risk of false signals.

Why does the S&P 500 lead the economy?

Equity markets are FORWARD-LOOKING. Investors price in expected corporate earnings 6–12 months ahead. When investors anticipate an economic recovery, they buy equities before GDP data improves.

This is why the S&P 500 is included in the US Conference Board Leading Economic Index (LEI).

Similarly, the yield curve inverts (short rates > long rates) when investors expect the Fed to cut rates — anticipating recession. Yield curve inversion has preceded every US recession since 1970.

US Leading Economic Indicators — CFA LM2, Exhibit 11

CFA Vol 2 LM2 presents the Conference Board US Leading Economic Index (LEI) components with rationale for each. These are heavily tested.

Leading Indicator	CFA LM2 Rationale
Avg weekly hours, manufacturing	Businesses cut overtime BEFORE laying off workers. Hours adjust first.
Initial jobless claims (weekly)	Sensitive early test of new layoffs. Rises before unemployment rate.
ISM new orders index	Month-on-month change in orders for final sales. Decline signals weak demand.
Manufacturers new orders (capital goods)	Captures business investment expectations. First signal of investment moves.
Building permits (new private housing)	Signals new construction activity. Permits required before building.
S&P 500 equity index	Stocks anticipate economic turning points. Forward-looking earnings proxy.
Yield curve spread (10yr Treasury – Fed Funds)	Inversion = market expects rate cuts = recession expected. Reliable predictor.
Leading Credit Index	Aggregates 6 financial indicators. Measures financial system fragility.
Consumer expectations index	Optimism raises spending. Provides early insight into household direction.

PMI, Big Data & Nowcasting — Modern Cycle Tracking

Traditional indicators have lags. Modern techniques use real-time data to track the economy as it happens — called nowcasting.

PMI > 50 = expansion **PMI < 50 = contraction** **PMI = 50 = no change**

Purchasing Managers Index (PMI) — A Key CFA Indicator

PMI surveys purchasing managers at hundreds of companies monthly on: new orders, output, employment, supplier delivery times and inventory.
A reading above 50 indicates more managers are seeing expansion than contraction. Below 50 = contraction.
Manufacturing PMI and Services PMI are reported separately. Composite PMI covers both.

PMI advantages: (1) Very timely — published within days of month-end. (2) Available across 30+ countries. (3) Based on actual business sentiment, not modelled data. (4) Strong lead time vs official GDP data.

CFA note: The Eurozone CLI includes Eurozone Manufacturing PMI and Services PMI as components.

Nowcasting (e.g. GDPNow)

The Atlanta Fed's GDPNow uses real-time data to estimate current-quarter GDP BEFORE official data is released. Updated as new monthly data flows in. Used by traders and central bankers.

Big Data Indicators

Credit/debit card transaction volumes, web search trends (Google Trends), mobility data (Apple, Google), satellite data on shipping/car park activity. All used to supplement official statistics.

SECTION 03

Fiscal Policy (CFA LM3)

Fiscal Policy — Roles and Objectives (CFA LM3)

Fiscal Policy = government spending + taxation decisions to influence aggregate demand

CFA LM3: fiscal policy has three roles in the economy. It is a powerful but slow-acting tool.

1. Manage aggregate demand

Expansionary policy (tax cuts, spending rises) stimulates demand. Contractionary (tax rises, spending cuts) dampens inflationary pressure. Keynesians favour fiscal intervention; Monetarists prefer monetary policy.

2. Redistribute income and wealth

Progressive taxes + transfer payments reduce inequality. Social spending on health, education, pensions. CFA notes this is secondary to stabilisation.

3. Allocate resources between sectors

Government spending shifts resources toward public goods (infrastructure, defence). Tax incentives direct private investment (R&D credits, green subsidies).

Keynesians vs Monetarists — The Fiscal Policy Debate (CFA LM3)

CFA LM3 presents both schools. The debate over fiscal multipliers has real consequences for policy design and asset allocation.

Keynesian View

Fiscal policy is POWERFUL when there is spare capacity. Multiplier > 1: GBP1 of government spending can generate > GBP1 of GDP growth (via second-round effects).

In a recession: tax cuts and spending rises boost demand directly. Aggregate demand can be stuck below potential.

Favour: countercyclical fiscal policy, deficit spending in recessions, automatic stabilisers.

Monetarist View

Fiscal policy has only TEMPORARY effect on demand. Crowding out: government borrowing raises interest rates, reducing private investment. Net effect limited.

Ricardian equivalence: consumers anticipate future tax rises to repay debt, so they save more now. Tax cuts fail to stimulate.

Favour: monetary policy over fiscal policy. Rules-based approaches. Independent central banks.

IMF Research on Fiscal Multipliers (CFA LM4 reference)

IMF study: with monetary accommodation, government expenditure multiplier = 3.9 over two years (vs 1.6 without accommodation).
Targeted social transfers to the poorest: multiplier = 1.7 with monetary accommodation (vs 0.5 without).
Conclusion: fiscal multipliers are LARGER when monetary policy is also loose (lower interest rates reinforce fiscal stimulus).

Government Revenue & Expenditure — CFA LM3 Data

CFA LM3 provides data on general government revenues and expenditures as % of GDP across major economies. Context for fiscal space analysis.

Country	Revenue % GDP (1995)	Revenue % GDP (2010)	Revenue % GDP (2015)	Fiscal Space vs Germany
Australia	34.5%	32.4%	34.9%	More space (lower tax burde
Germany	45.1%	43.0%	44.5%	Benchmark — high revenue
Japan	31.2%	30.6%	35.7%	More space
United Kingdom	38.2%	38.2%	38.0%	Moderate
United States	33.8%	30.9%	33.4%	More space (lower taxes rela spending)
OECD average	37.9%	39.8%	40.9%	Trending up post-2008

Automatic Stabilisers vs Discretionary Fiscal Policy

Automatic stabilisers: built-in fiscal mechanisms that act WITHOUT legislative action. Unemployment benefits rise in recession (more spending automatically). Taxes fall when incomes fall (less revenue automatically). Both cushion demand without policy delays. Discretionary policy: deliberate tax changes or spending programmes. CFA notes this suffers from recognition, legislative and implementation lags — may take 1–2 years to impact the economy.

Fiscal Deficit, National Debt & Investment Implications

Primary Deficit = G − T (excl. interest) **Overall Deficit = Primary Deficit + Interest Payments**

The debt-to-GDP ratio is the key metric for assessing fiscal sustainability. CFA LM3 asks: does it matter how large the national debt is?

Argument	Position	CFA LM3 Assessment
Debt does not matter	Ricardian equivalence: private saving rises to offset deficit. No crowding out. Future taxes = current debt.	Theoretical but empirically weak. Individuals do not fully offset fiscal changes.
Debt DOES matter	Crowding out: government borrowing competes with private sector for funds, raising interest rates. Reduces private investment.	More empirical support, especially in small open economies or near full capacity.
Debt sustainability	r < g: if growth rate > interest rate, debt/GDP ratio shrinks naturally without primary surplus.	Used by IMF/ECB to assess sustainability. Fragile if interest rates rise.

Bond investor implication: sovereign yields reflect fiscal credibility. High debt + political dysfunction = higher risk premium (e.g. peripheral Eurozone spreads in 2012 crisis).

SECTION 04

Monetary Policy (CFA LM4)

Money, Central Banks & Policy Tools (CFA LM4)

Monetary policy: central bank influences quantity of money and credit to affect macroeconomy

CFA LM4: monetary policy operates through interest rate channels, credit channels and asset price channels.

Tool	How It Works	Expansionary (Cut/Increase?)	Tightening (Cut/Increase?)
Policy rate (e.g. Fed Funds)	Sets base short-term borrowing cost. Cascades to all credit rates.	Cut rates → cheaper borrowing → more spending	Raise rates → dearer borrowing → less spending
Reserve requirements	% of deposits banks must hold. Affects money multiplier.	Lower requirement → banks can lend more	Raise requirement → banks lend less
Open market operations	Buy (sell) government bonds to inject (drain) bank reserves.	Buy bonds = inject reserves = more credit	Sell bonds = drain reserves = less credit
Forward guidance	Signal future policy direction. Manages expectations.	Promise low rates for long time = stimulative now	Signal coming hikes = tightens expectations
Quantitative easing (QE)	Buy long-dated bonds/assets when short rates hit zero (ZLB).	Compress long-term yields. Portfolio rebalancing.	Quantitative tightening (QT) = reverse QE

Neutral Rate, Inflation Targeting & Policy Stance (CFA LM4)

Neutral (Natural) Rate r* = rate that neither stimulates nor slows the economy. Consistent with stable long-run inflation.

CFA LM4: the neutral rate is the benchmark for judging whether monetary policy is EASY or TIGHT.

Policy rate < neutral rate	EASY/ACCOMMODATIVE monetary policy. Stimulates demand and credit growth. Output gap expected to close. Appropriate in recession/recovery phase.
Policy rate = neutral rate	NEUTRAL monetary policy. Neither adding nor removing stimulus. Appropriate when economy is at potential with stable inflation.
Policy rate > neutral rate	TIGHT/RESTRICTIVE monetary policy. Slows demand and credit. Appropriate when economy is overheating or inflation is above target.

Inflation target: most major central banks target 2% CPI inflation. Taylor Rule approximation: policy rate = neutral rate + 0.5(inflation gap) + 0.5(output gap).

Monetary-Fiscal Policy Mix — CFA LM4 Combinations

CFA LM4 explicitly analyses four combinations of monetary and fiscal policy stance. Each has different effects on growth, rates and portfolio composition.

Monetary Policy	Fiscal Policy	Economic Effect	Portfolio Implication
Easy	Easy	Highly expansionary. Rising demand. Lower real rates (if monetary eases more). Growing public AND private sector.	Equities, inflation assets, commodities. Duration short.
Easy	Tight	Private sector stimulus. Public sector contracts. Lower interest rates. Private investment grows as share of GDP.	Equities (especially capex-driven). Bonds benefit from low rates.
Tight	Easy	Higher interest rates due to fiscal stimulus. Government spending rises as share of GDP. Some crowding out of private investment.	Short bonds. Government bonds underperform. Defensive equities.
Tight	Tight	Sharply contractionary. Interest rates rise (if monetary impact dominant). Demand falls from both public and private.	Long government bonds if deflationary. Cash. Avoid equities.

Quantitative Easing (QE) — When Rates Hit Zero

When the policy rate reaches zero (zero lower bound, ZLB), conventional monetary policy is exhausted. QE is the primary unconventional alternative.

How QE Works

Central bank creates new bank reserves and uses them to BUY long-dated government bonds (and sometimes corporate bonds or mortgage-backed securities) from financial institutions.

Channels of impact:
1. Reduce long-term yields: buying bonds pushes prices up, yields down. Lowers borrowing costs for businesses and households.
2. Portfolio rebalancing: investors who sell bonds to the central bank redeploy into riskier assets (equities, corporate bonds), compressing spreads and raising asset prices.
3. Exchange rate: lower yields make domestic currency less attractive to foreign investors. Currency depreciates, supporting exports.
4. Signalling: commitment to QE signals rates will remain low, anchoring expectations.

QE in Practice

Fed QE1 (2008): USD 1.75tn. Fed QE3 (2012): USD 85bn/month.
ECB PSPP (2015): EUR 60bn/month. BoE (2009-2020): GBP 450bn.
2020 pandemic: coordinated global QE of unprecedented scale.

Risks and Limits

Asset price inflation: QE benefits holders of financial assets, widening inequality.
Inflation risk if central bank is too slow to unwind (2021-22 inflation surge).
Reverse QE (quantitative tightening/QT): sell bonds back, reduce balance sheet.

Inflation, Deflation & Stagflation — Investment Implications

CPI inflation = % change in cost of fixed consumer basket **Core inflation = excl. food and energy**

Inflation is the most important macroeconomic variable for fixed income investors. It determines REAL returns and central bank policy.

Scenario	Definition	Central Bank Response	Asset Impact
Moderate inflation (1–3%)	'Goldilocks': consistent with central bank targets. Healthy demand.	Neutral to slight tightening.	Equities, real estate, commodities all positive.
High inflation (> 3–4%)	'Overheating': demand exceeds potential. Often wage-price spiral risk.	Aggressive tightening. Rate rises.	Bonds suffer (prices fall as yields rise). Real assets outperform.
Deflation (< 0%)	Falling prices. Consumers delay purchases. Debt burden rises in real terms.	Aggressive easing. QE. Near-zero rates.	Long government bonds outperform. Equities weak.
Stagflation	High inflation + high unemployment simultaneously. Rare. Supply shocks (e.g. 1970s oil).	No easy response: tightening worsens unemployment.	Commodities (especially oil) outperform. Both bonds and equities suffer.

TIPS (Treasury Inflation-Protected Securities): principal rises with CPI. Real yield = nominal yield − breakeven inflation. When inflation rises above breakeven, TIPS outperform nominal bonds.

Unemployment — Types, Measurement & Policy Relevance

Unemployment rate = unemployed / labour force × 100

Labour force = employed + actively seeking work

Unemployment is a key LAGGING indicator. It peaks after the recession has ended and only falls well into recovery. Used to confirm cycle phase.

Type	Definition	Policy Fix?	Investment Relevance
Frictional	Voluntary unemployment while searching for a better job.	No: healthy market mobility	Low. Normal even in healthy economy.
Structural	Skills mismatch: jobs exist but workers lack required skills.	Yes: education, retraining	Medium. Signals long-term labour market issues.
Cyclical (demand-deficient)	Insufficient aggregate demand. Recession-driven layoffs.	Yes: fiscal/monetary stimulus	High. Rises sharply in contraction. Target of macro policy.
Natural rate (NAIRU)	Frictional + structural. Below this = inflationary labour markets.	No: equilibrium concept	Fed target. If unemployment < NAIRU, inflation likely rises.

NAIRU (Non-Accelerating Inflation Rate of Unemployment): central banks cannot push unemployment below NAIRU without triggering persistent inflation. Key for wage/price spiral risk.

SECTION 05

International Trade (CFA LM6)

Comparative Advantage — The Foundation of Trade

Trade based on COMPARATIVE advantage (lowest opportunity cost) benefits all parties even if one is better at everything

David Ricardo (1817): countries should specialise in what they can produce at the LOWEST RELATIVE COST, not just lowest absolute cost.

WORKED EXAMPLE: UK and Portugal

Without trade: UK needs 100 labour units for wine, 90 for cloth. Portugal needs 80 for wine, 120 for cloth.
UK has ABSOLUTE advantage in cloth (90 < 120). Portugal has absolute advantage in wine (80 < 100).

Opportunity cost of wine: UK = 100/90 = 1.11 units of cloth. Portugal = 80/120 = 0.67 units of cloth.
Opportunity cost of cloth: UK = 90/100 = 0.90 units of wine. Portugal = 120/80 = 1.50 units of wine.

UK has comparative advantage in cloth (0.90 < 1.50 units of wine forgone).
Portugal has comparative advantage in wine (0.67 < 1.11 units of cloth forgone).
Both gain by specialising and trading even though UK is better at cloth in absolute terms.

Free trade maximises global output. But benefits are not evenly distributed — import-competing sectors lose, creating political pressure for protection.

Tariff Analysis — CFA LM6: South Africa Paper Industry

Directly from CFA Vol 2 LM6, Example 1. A complete welfare analysis of a 20% import tariff on paper. The definitive CFA trade example.

CFA LM6 EXAMPLE 1 — SOUTH AFRICA PAPER TARIFF

World price = USD 5.00/ton. South Africa domestic production = 110,000 tons. Domestic demand = 200,000 tons. Imports = 90,000 tons.

20% tariff imposed: new price = USD 6.00/ton. Domestic production rises to 130,000 tons. Demand falls to 170,000 tons. Imports fall to 40,000 tons.

CFA Calculations:

- Consumer surplus LOSS = USD1 × 170,000 + ½ × USD1 × 30,000 = USD 185,000 (areas A+B+C+D)
- Producer surplus GAIN = USD1 × 110,000 + ½ × USD1 × 20,000 = USD 120,000 (area A)
- Government revenue = USD1 × 40,000 = USD 40,000 (area C)
- Deadweight loss = ½ × USD1 × 20,000 + ½ × USD1 × 30,000 = USD 25,000 (areas B+D)

Net welfare effect = −185,000 + 120,000 + 40,000 = −25,000 = deadweight loss.

CFA rule: in a SMALL country, a tariff always reduces national welfare by the deadweight loss amount. Consumer surplus loss = producer gain + government revenue + DWL.

Trade Restrictions — Tariffs, Quotas, Subsidies (CFA LM6)

CFA LM6 provides a detailed comparison of four trade restriction types. Tariffs are the most investor-relevant (tariff wars, WTO disputes).

Impact on	Tariff	Import Quota	Export Subsidy	VER
Who bears it	Importing country	Importing country	Exporting country	Importing country
Domestic price	Increases	Increases	Increases (domestic)	Increases
Producer surplus	Increases	Increases	Increases	Increases
Consumer surplus	Decreases	Decreases	Decreases (domestic)	Decreases
Government revenue	Increases	Mixed (quota rent)	Decreases (cost)	No change
National welfare	Decreases (small country)	Decreases	Decreases	Decreases
Key difference	Govt captures rent	Exporters may capture rent	Subsidises competitors' customers	Exporter sets limit

SECTION 06

Exchange Rates (CFA LM7)

Exchange Rate Conventions — CFA LM7

Direct quote: domestic currency / foreign currency (cost of 1 unit of foreign currency in domestic terms)

CFA LM7 is precise about quoting conventions. Errors in quoting lead to errors in calculating appreciation/depreciation.

CFA LM7, Example 3 — MXN/USD Quote

A dealer in New York provides a spot quote of 18.8590 MXN/USD to a client in Mexico City.
From the MEXICAN client perspective: the quote is MXN/USD = 18.8590.
Direct quote (domestic/foreign) for a Mexican: MXN is domestic, USD is foreign. MXN18.8590 per USD = DIRECT quote. CFA answer: B.
Inverse: 1/18.8590 = USD 0.0530 per MXN (indirect quote from Mexican perspective, or direct from US perspective).

Bid/offer (18.8580–18.8600 MXN/USD):
In USD/MXN terms: bid = 1/18.8600 = 0.05302, offer = 1/18.8580 = 0.05303 (bid < offer always). CFA answer: A.

Direct quote

Price currency = domestic currency. Foreign currency is the base (1 unit of foreign = X domestic).
UK investor: GBP/USD = 1.25 means GBP1.25 per USD1 (direct).

Indirect quote

Price currency = foreign currency. Domestic is the base (1 unit of domestic = X foreign).
US investor: USD/GBP = 0.80 means USD0.80 per GBP1 (indirect for UK investor).

Currency Appreciation — CFA LM7: USD/EUR Worked Example

Directly from CFA Vol 2 LM7. Calculating percentage appreciation/depreciation requires care about which currency is base and which is price.

% change = (New Rate / Old Rate) − 1 [always measure change in the PRICE currency relative to the BASE]

CFA LM7: USD/EUR rises from 1.1500 to 1.2000
USD/EUR = 1.1500 at t=0. USD/EUR = 1.2000 at t=1. (USD is PRICE currency, EUR is BASE currency.)

Euro (base) appreciation vs USD (price):
% Δ = 1.2000 / 1.1500 − 1 = 4.35% ← CFA ANSWER (euro appreciates 4.35% vs USD)

Dollar depreciation vs euro:
Invert: at t=0: EUR/USD = 1/1.1500 = 0.8696. At t=1: EUR/USD = 1/1.2000 = 0.8333.
% Δ = 0.8333 / 0.8696 − 1 = −4.17% (USD depreciates 4.17%)

CRITICAL: euro APPRECIATION (+4.35%) ≠ dollar DEPRECIATION in % terms (−4.17%). These are NEVER equal.
The 4.35% vs 4.17% difference is a standard CFA exam trap.

CFA rule: percentage appreciation of base currency ≠ percentage depreciation of price currency (except when changes are very small).

Purchasing Power Parity & Interest Rate Parity

Two fundamental equilibrium frameworks for exchange rates. PPP is a long-run theory; IRP is a no-arbitrage short-run condition.

Purchasing Power Parity (PPP)

Law of one price extended to all goods:
$E(S_t) / S_0 = (1 + \text{inflation}^{\text{mdresterster}}) / (1 + \text{inflation}^{\text{fore}}{}_{i}{}^{\text{kn}})$

High inflation country's currency depreciates.
Big Mac Index: popular PPP proxy.
Works long-run; poor short-run predictor.

Interest Rate Parity (IRP)

No-arbitrage condition in FX forward markets:
$F/S = (1 + r^{dom}) / (1 + r^{for})$ (covered IRP)

Higher interest rate country's currency trades at forward DISCOUNT.
Uncovered IRP: expected spot = forward rate.
Violated by risk premium, capital controls, or carry trades.

Carry Trade and IRP

Carry trade: borrow in low-rate currency, invest in high-rate currency. Profits IF the high-rate currency does not depreciate as much as IRP predicts.
IRP violation: high-rate currencies often appreciate short-run (capital inflows attracted by higher yield). Carry trade profits from this deviation.
Risk: sudden reversal ('carry crash') when risk appetite falls. All positions unwind simultaneously. Classic example: Japanese yen carry trade 2008.

Exchange Rate Regimes — Fixed, Floating & Managed (CFA LM7)

The choice of exchange rate regime involves trade-offs between exchange rate stability, capital mobility and monetary policy independence.

Regime	Description	Monetary Policy Independence?	Example
Currency union / Dollarisation	Use another country's currency. No exchange rate risk.	NONE. Completely surrendered.	Eurozone, Ecuador (USD), Kosovo (EUR)
Currency board	Fixed 1:1 to anchor currency. 100% reserves backing.	NONE effectively (must match anchor policy).	Hong Kong (HKD/USD), Bulgaria (EUR)
Fixed peg	Central rate fixed. Defended with intervention.	LIMITED. Must defend peg rate.	Saudi Arabia (USD), many Gulf states
Managed float (dirty float)	Market-determined but CB intervenes to smooth.	PARTIAL. Can set own inflation target.	China, India, Singapore (basket peg)
Free float	Market-determined. No intervention.	FULL. Completely independent.	USD, EUR, GBP, JPY, AUD

Formula Reference Card — Session 4 Macroeconomics

Key formulas and relationships from this session.

GDP (Expenditure)	GDP = C + I + G + NX
Real GDP	Real GDP = Nominal GDP / GDP Deflator × 100
Output Gap	Output Gap = Actual GDP − Potential GDP
Unemployment Rate	= Unemployed / Labour Force × 100
CPI Inflation	% change in cost of fixed consumer basket year-on-year
Fisher Equation	$(1+r^{nomgenaL}) = (1+r^{reaL}) \times (1+\text{inflation})$ Approx: $r^{reaL} = r^{nomgenaL} - \text{inflation}$
Tariff consumer surplus loss	CS loss = $\Delta P \times Q^{neW} + ½ \times \Delta P \times \Delta Q^{deMand}$
Tariff government revenue	Govt revenue = tariff (ΔP) × quantity imported after tariff
Deadweight loss (tariff)	DWL = $½ \times \Delta P \times \Delta Q^{LnɔWL} + ½ \times \Delta P \times \Delta Q^{deMand}$
FX % appreciation	= New Rate / Old Rate − 1 (change in PRICE currency vs BASE currency)
PPP (relative)	$E(S_t)/S_0 = (1+\text{inflation}^{dom}) / (1+\text{inflation}^{for})$
Covered IRP	$F/S = (1+r^{dom}) / (1+r^{for})$ [F and S in domestic/foreign price terms]
Taylor Rule (approx)	Policy rate ≈ neutral rate + 0.5(inflation−target) + 0.5(output gap)

PRACTICE PROBLEMS

Work through each before checking answers.

Q1 Business cycle (CFA LM2): The ISM new orders index falls below 50 for three consecutive months while building permits also decline. What business cycle phase does this suggest and why?

Q2 Tariff (CFA LM6 South Africa): World price = USD 5. Tariff 20% → price = USD 6. Domestic production: 110k→130k tons. Demand: 200k→170k tons. Calculate: (a) consumer surplus loss, (b) deadweight loss.

Q3 Exchange rate (CFA LM7): USD/EUR rises from 1.0500 to 1.1025. What is the percentage appreciation of the euro? What is the percentage depreciation of the dollar?

Q4 Fiscal policy: An economy has an output gap of −2% of GDP. Government cuts income tax by 1% of GDP. Keynesian multiplier is 1.5. How much does GDP rise? Does this close the output gap?

Q5 Monetary-fiscal mix (CFA LM4): A central bank tightens policy while the government runs a large deficit (expansionary fiscal). What are the likely effects on (a) GDP growth, (b) interest rates, (c) private investment?

Practice — Worked Solutions

A1

ISM <50 = contraction in manufacturing orders. Declining building permits = forward-looking weakness in construction. Together these are LEADING indicators signalling the economy is entering or already in a CONTRACTION phase. Output gap is widening negatively. Central bank likely to ease policy. Investors should shift toward bonds and defensive equities.

A2

(a) Consumer surplus loss = USD1 × 170,000 + ½ × USD1 × 30,000 = USD 170,000 + USD 15,000 = USD 185,000 (CFA answer). (b) Deadweight loss = ½ × USD1 × 20,000 + ½ × USD1 × 30,000 = USD 10,000 + USD 15,000 = USD 25,000. Note: producer gain = USD 120,000, government revenue = USD 40,000. Check: 185k = 120k + 40k + 25k = 185k. ✓

A3

Euro appreciation: 1.1025/1.0500 − 1 = 5.00%. Dollar depreciation: inverse at t0 = 1/1.0500 = 0.9524. At t1 = 1/1.1025 = 0.9070. Change = 0.9070/0.9524 − 1 = −4.76%. CFA rule: 5.00% ≠ 4.76%. These percentages are never equal. The trap: 1/1.05 = 4.76%, not 5%.

A4

GDP rise = multiplier × fiscal injection = 1.5 × 1% = 1.5% of GDP. Output gap = −2%. Fiscal stimulus closes 1.5/2 = 75% of the gap. Not fully closed. Need additional stimulus or time for automatic stabilisers. Limitation: multiplier may be lower if consumers save the tax cut (Ricardian equivalence, leakages).

A5

(a) GDP growth: mixed effects. Tight monetary restrains private demand; loose fiscal stimulates government demand. Net effect depends on relative magnitudes. (b) Interest rates: likely HIGHER. Government borrowing competes with private sector for funds. CFA: crowding out risk. (c) Private investment: likely falls. Higher rates raise cost of capital. Government spending displaces private investment (crowding out). Composition of GDP shifts toward public sector.

KEY TAKEAWAYS

Session 4 — Macroeconomics

- 4 CFA phases (LM2): Recovery (neg. gap narrows), Expansion (positive gap), Slowdown (positive gap narrows), Contraction (neg. gap widens). Asset rotation by phase.
- US leading indicators (CFA LM2): hours, jobless claims, ISM orders, building permits, S&P 500, yield curve spread, Credit Index. All lead by 3–12 months.
- South Africa tariff (CFA LM6): consumer surplus loss = USD 185k. Producer gain = 120k. Govt revenue = 40k. DWL = 25k. Small country always loses net welfare.
- USD/EUR: 1.1500 → 1.2000 = +4.35% euro appreciation (not +4.17%). CFA exam trap: appreciate ≠ depreciate in % terms.
- MXN/USD = 18.8590 is direct quote for Mexican client (domestic MXN / foreign USD).
- Monetary-fiscal mix (CFA LM4): easy fiscal + tight monetary = higher rates, crowding out. Easy both = highly expansionary. Know all 4 combinations.

- Keynesian: fiscal multiplier >1. Monetarist: crowding out +
- PPP: high inflation country currency depreciates. IRP: high